THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Real Numbers, Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The prime factorisation of 140 is: 1

(a) $2 \times 5 \times 7$ **(b)** $2^3 \times 5 \times 7$ **(c)** $2^2 \times 5 \times 7$ **(d)** $2^2 \times 3 \times 7$

2. The HCF of 8, 9 and 25 is: 1

(a) 9 **(b)** 1 **(c)** 8 **(d)** 25

3. If $\text{HCF}(90, 144) = 18$, then $\text{LCM}(90, 144)$ is: 1

(a) 720 **(b)** 540 **(c)** 360 **(d)** 1440

4. A student states: "$\sqrt{9}$ is irrational because it has a root sign." The correct response is: 1

(a) Correct for odd numbers **(b)** Correct

(c) Incorrect; $\sqrt{9} = 3$ is rational **(d)** Cannot be determined

5. For the LCM of 15 and 25, Student A gets 75 and Student B gets 375. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The zeroes of $x^2 + 2x - 15$ are: 1

(a) $3, -5$ **(b)** $15, -1$ **(c)** $-3, 5$ **(d)** $5, -3$

7. The graph of $y = p(x)$ crosses the x-axis at exactly 4 points. The number of zeroes of $p(x)$ is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 0

8. If α, β are the zeroes of $x^2 - 8x + 15$, then $\alpha + \beta + \alpha\beta$ is: 1

(a) 120 **(b)** 23 **(c)** 7 **(d)** −7

9. A quadratic polynomial whose zeroes are 4 and -1 is: 1

(a) $x^2 - 3x - 4$ **(b)** $x^2 - 5x + 4$ **(c)** $x^2 + 3x + 4$ **(d)** $x^2 + 3x - 4$

10. A student states: "The degree of the polynomial 7 is 1." The correct response is: 1

(a) Correct

(b) It has no degree

(c) Its degree is 7

(d) Incorrect — a non-zero constant has degree 0

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the HCF and LCM of 24 and 36 by prime factorisation and verify that HCF $\times$ LCM $=$ product of the numbers. **2**

12. Show that $\frac{2\sqrt{3}}{5}$ is irrational, given that $\sqrt{3}$ is irrational. **2**

13. Find the zeroes of $3x^2 - 75$ and verify the relationship between the zeroes and the coefficients. **2**

14. If one zero of $2x^2 + 3x + k$ is $\frac{1}{2}$, find k and the other zero. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{11}$ is irrational. **3**

16. Find the zeroes of $\sqrt{3}x^2 - 8x + 4\sqrt{3}$ and verify the relationship between the zeroes and the coefficients. **3**

17. Find the greatest number of children among whom 1001 biscuits and 182 chocolates can be equally distributed. How many of each does a child get? **3**

OR

Two numbers are in the ratio 3 : 4 and their LCM is 180. Find the numbers. **3**

18. If α, β are the zeroes of $x^2 - 3x + 2$, find a quadratic polynomial whose zeroes are $\alpha + 2$ and $\beta + 2$. **3**

OR

If the sum of the squares of the zeroes of $x^2 - 8x + k$ is 40, find k. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

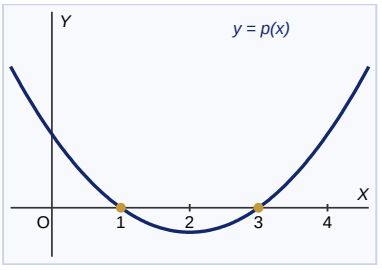
19. **Case study:** A school has 312 students in Class IX and 260 in Class X. They are to be divided into teams of equal size, each team having students of one class only. **5**

(i) Find the maximum size of a team.
(ii) Find the total number of teams.
(iii) Can 208 Class VIII students also be made into teams of the same size? Justify.

OR

(i) Prove that $\sqrt{2}$ is irrational. **5**
(ii) Hence prove that $\sqrt{2} + \sqrt{3}$ is irrational.

20. **Case study:** The graph of a quadratic polynomial $y = p(x)$ is shown. **5**



(i) Write its zeroes.
(ii) Write $p(x)$ if its leading coefficient is 1.
(iii) Verify the relationship between the zeroes and coefficients.
(iv) Find $p(2)$.

OR

Find the zeroes of $p(x) = 2x^2 - 9 - 3x$ and verify the relationship between the zeroes and coefficients. Also find a polynomial whose zeroes are the negatives of these zeroes. **5**

END OF QUESTION PAPER — **Total: 40 Marks**